DATE : 20.10.2026

DT/NT : DT

LESSON : AWS

SUBJECT: S3-1

BATCH : B 424-020DE

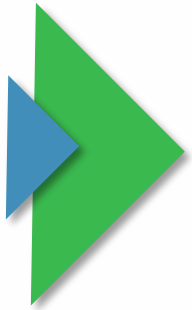
AWS-DEVOPS

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Table of Contents

- Introduction to S3
- Buckets & Objects
- Storage Classes
- Versioning
- Secure Website Hosting

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Introduction to S3

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Introduction to S3

What is S3?

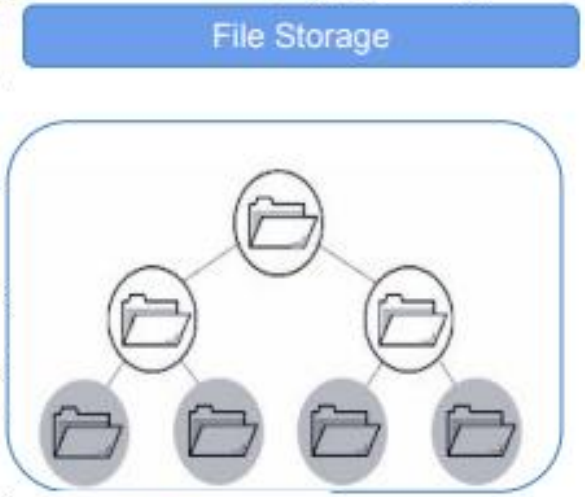
Amazon Simple Storage Service

- Object storage for files, backups, static assets and data lakes.
- Data is stored as objects inside buckets in an AWS Region.
- S3 is managed by AWS; you do not manage storage servers.
- Access is private by default unless granted explicitly.

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Introduction to S3

Storage Options



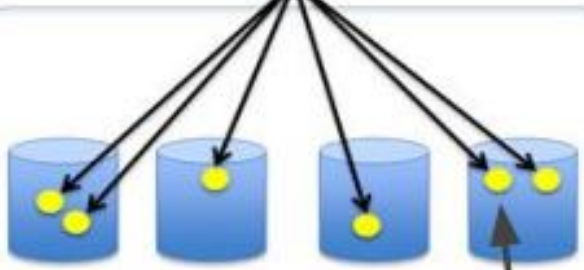
File Storage

Amazon EFS

- Store virtually unlimited files.
- Maintain file revisions.
- HTTP(S) based interface.
- Files are distributed in different physical nodes.

HTTP(S) Interface

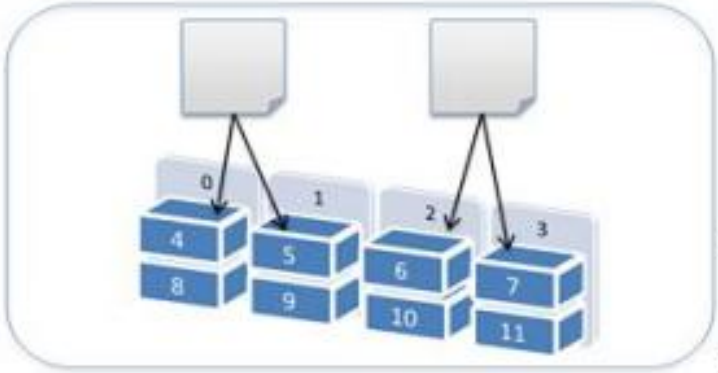
Object Storage



Object=
File+
Unique ID+
Metadata+

Amazon Elastic Block Storage (EBS)

Block Storage

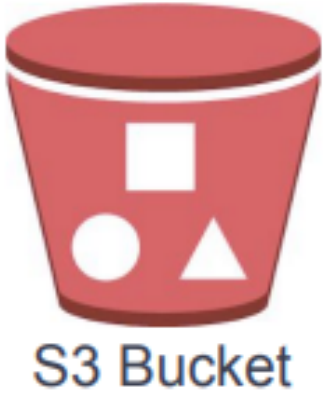


- File is split and stored in fixed sized blocks.
- Capacity can be increased by adding more nodes.
- Suitable for applications which require high IOPS, database, transactional data.

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Introduction to S3

What is S3 Bucket?



- A bucket is a logical storage unit used to store objects in AWS.
- A bucket can also be considered as a container.

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Introduction to S3

S3 Buckets and Limits

How Buckets Work

- A general purpose bucket stores any number of objects.
- Default quota: 10,000 general purpose buckets per AWS account (quota increases may be requested).
- Bucket names must meet AWS naming rules and be unique within the relevant namespace.
- The bucket Region is selected at creation.

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Introduction to S3

Region, Durability and Availability

Region and Names

- Select an AWS Region for each bucket.
- Bucket names follow AWS uniqueness rules.
- Data stays in that Region unless copied or replicated.

Availability Zones

- S3 Standard uses multiple AZs.
- S3 One Zone-IA and S3 Express One Zone store data in one AZ.
- Durability and availability are different metrics.

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Introduction to S3

S3 Object Size

Maximum Object Size

Up to 50 TB per object.

For large objects, use multipart upload with AWS CLI, SDK, or API.

AWS Console Upload

Up to 160 GB per uploaded object in the S3 Console.

Objects have a key, data, metadata and optional tags.

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Introduction to S3

Uploading S3 Objects

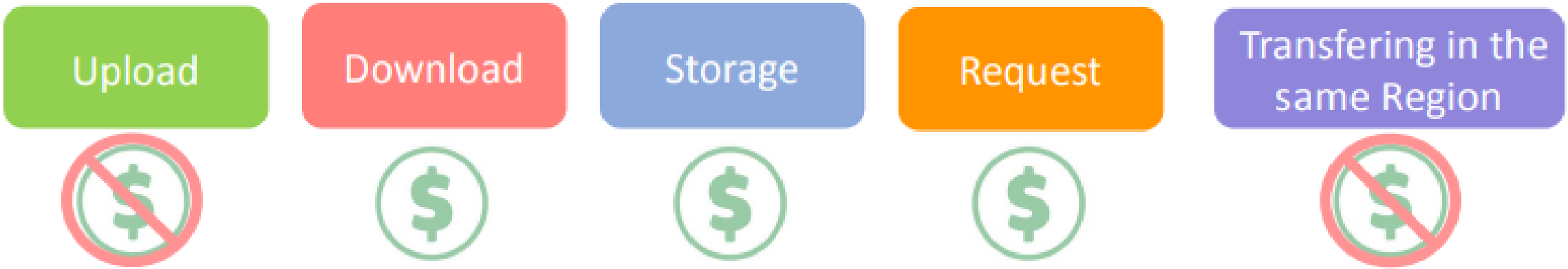
Choose Your Upload Method

- Console uploads: convenient for classroom files, up to 160 GB per object.
- AWS CLI / SDK / API: automation and multipart upload for larger files.
- Block Public Access: keep enabled by default.
- Object Ownership: prefer Bucket owner enforced (ACLs disabled).

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Introduction to S3

S3 Object Pricing



Storage Classes

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Storage Classes

Current S3 Storage Classes

RRS still exists but AWS does not recommend it for new designs.

Storage Class	Typical Need	Access Pattern
S3 Standard	Frequently used files	Milliseconds
Intelligent-Tiering	Unknown/changing access	Auto cost tiers
Standard-IA	Infrequent, multi-AZ	Milliseconds
One Zone-IA	Infrequent, one AZ	Milliseconds
S3 Express One Zone	Latency-sensitive / single AZ	Single-digit ms
Glacier Instant Retrieval	Rarely accessed archive	Milliseconds
Glacier Flexible / Deep Archive	Long-term archives	Minutes to hours

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Storage Classes

S3 Standard

Default Storage Class

- Used for frequently accessed data with low-latency access.
- Designed for 99.999999999% (11 nines) durability.
- Designed for 99.99% availability.
- Stores data redundantly across multiple Availability Zones.
- Good for web assets, business data, and active applications.

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Storage Classes

S3 Express One Zone

When to Use It

- Very low-latency, high-request workloads.
- Data is stored within one Availability Zone.
- Uses S3 directory buckets, not general purpose buckets.

What About RRS?

- Reduced Redundancy Storage is not recommended by AWS.
- S3 Standard is typically more cost-effective.
- RRS is a legacy concept, not a new-project choice.

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Storage Classes

S3 Standard-Infrequent Access

Suitable for

- Data needed immediately but read infrequently.
- Stored across multiple Availability Zones.
- Common for disaster recovery files and backup copies.

Pricing Conditions

- 30-day minimum storage duration.
- 128 KB minimum billable object size.
- Retrieval charges may apply.

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Storage Classes

S3 One Zone-Infrequent Access

Design Choice

- Lower-cost, single-AZ storage.
- Rapid access for infrequently used data.
- No protection against the loss of the whole AZ.

Conditions

- 30-day minimum storage duration.
- 128 KB minimum billable size.
- Suitable for reproducible or secondary data, not your only critical copy.

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Storage Classes

S3 Intelligent-Tiering

Three Automatic Access Tiers

- Frequent Access
- Infrequent Access
- Archive Instant Access
- All support millisecond access.

Optional Archive Tiers

- Archive Access and Deep Archive Access can be enabled.
- Archival tiers require asynchronous retrieval.
- Monitoring/automation charges apply to eligible objects.

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Storage Classes

S3 Glacier Storage Classes

Instant Retrieval

- Archive data accessed rarely.
- Millisecond retrieval.
- 90-day minimum storage duration.

Flexible Retrieval

- Retrieval in minutes to hours.
- Restore required.
- 90-day minimum duration.

Deep Archive

- Lowest-cost long-term archives.
- Hours to retrieve.
- 180-day minimum duration.

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Storage Classes

S3 Glacier Deep Archive

Long-Term Archives

- For data accessed very rarely, such as compliance archives.
- Minimum storage duration: 180 days.
- Typical retrieval is measured in hours (not milliseconds).
- Costs may include storage, retrieval requests and data retrieval.
- Not appropriate for a frequently accessed website.

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Storage Classes

Storage Classes — Fast Comparison

Class	Where Stored	Typical Access
Standard	Multiple AZs	Frequent, instant
Intelligent-Tiering	Multiple AZs	Changing, automatic
Standard-IA	Multiple AZs	Rare, instant
One Zone-IA	One AZ	Rare, instant
Express One Zone	One AZ, directory bucket	Very low latency
Glacier Instant	Multiple AZs	Rare, instant
Glacier Flexible	Multiple AZs	Restore needed
Deep Archive	Multiple AZs	Long restore

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Storage Classes

Choosing a Storage Class

Need instant access?

Yes → S3 Standard, Standard-IA, Intelligent-Tiering or Express One Zone.

Consider frequency and architecture.

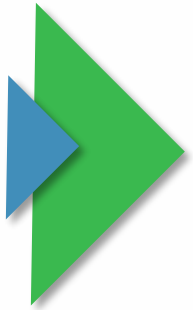
Access unpredictable?

Choose Intelligent-Tiering when its features and monitoring costs fit the workload.

Long-term archive?

Glacier Instant, Flexible or Deep Archive depending on retrieval urgency and retention.

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Versioning

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Versioning

What is S3 Versioning?

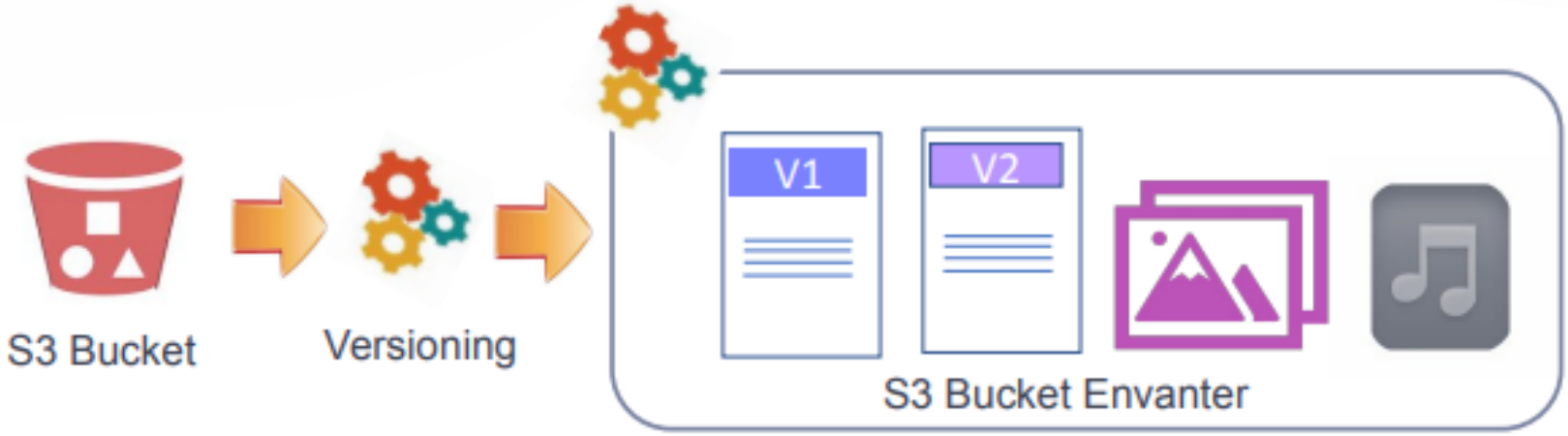
Protect Object History

- Versioning stores multiple full versions of an object in one bucket.
- Uploading the same key creates a new version.
- A normal delete creates a delete marker; older versions remain.
- Permanently deleting a specific version cannot be undone.
- Each stored version contributes to storage charges.

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Versioning

What is Versioning?



- Versioning is bucket-based feature

Versioning

S3 Versioning States

Three States of Versioning

- Unversioned (default): no enabled version history.
- Enabled: new uploads receive unique version IDs.
- Suspended: versioning stops assigning new unique IDs to uploads; old versions remain.
- After versioning has been enabled, you cannot return the bucket to its original unversioned state.

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S3 Static Website Hosting

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S3 Static Website Hosting

S3 Static Website Hosting

Static Content Only

- Hosts HTML, CSS, images and client-side JavaScript.
- It does not execute server-side code such as PHP or Node.js.
- Native S3 website endpoints support HTTP, not HTTPS.
- For production HTTPS, prefer private S3 content with CloudFront OAC.

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S3 Static Website Hosting

Static Website Access Methods

Recommended: CloudFront + OAC

- Keep the S3 bucket private.
- Block Public Access stays enabled.
- CloudFront delivers HTTPS to viewers.
- OAC authorizes CloudFront to read objects.

Optional Public Lab

- An isolated test bucket may use the native S3 website endpoint.
- Public reads need careful bucket policy settings.
- Website endpoint supports HTTP only.

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S3 Static Website Hosting

Three Different Ways to Open an Object

S3 Object URL

Direct object URL is denied while the object is private.

Public web access is not automatic.

Presigned URL

Temporary, signed access to one object.

Useful for private sharing.

CloudFront URL

HTTPS distribution serves approved S3 content securely with OAC.

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S3 Static Website Hosting

Public Access and ACLs

ACLs Disabled by Default

- Prefer Object Ownership: Bucket owner enforced.
- ACLs are disabled.
- Use IAM policies, bucket policies and CloudFront OAC.

Do Not Make Every Object Public

- Block Public Access ON by default.
- Do not disable account-wide protection for a lab.
- Avoid “Make public using ACL” instructions.

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S3 Static Website Hosting

Bucket Policies vs. ACLs

Bucket Policy

- JSON resource policy attached to the bucket.
- Can grant narrowly scoped s3:GetObject for an approved public lab.
- An explicit deny still takes precedence.

Object ACL

- A legacy object-level mechanism.
- Disabled for Bucket owner enforced buckets.
- Not necessary for the recommended lessons.

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S3 Static Website Hosting

Static Website Steps

1. Upload

Create a private bucket; upload index.html, error.html and assets/ with correct object keys.

2. Publish securely

Create a CloudFront distribution using the bucket REST origin and Origin Access Control.

3. Verify

Grant CloudFront narrow access, set default root object, verify HTTPS and cleanup.

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S3 Static Website Hosting

Native S3 Website Endpoint

Public Demo: Explicit Opt-in

- Properties → Static website hosting → Enable.
- Supply index.html and error.html.
- Bucket-level public read policy only for demo content.
- Endpoint is HTTP only.

Security Warning

- AWS recommends CloudFront for public websites.
- Do not use public buckets for secrets.
- Account / organization restrictions may prohibit public access.

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Versioning

Delete Markers and Recovery

Overwrite a File

Upload index.html twice → two stored versions.

The newest version is shown by default.

Delete and Restore

A normal delete adds a delete marker.

Remove that specific marker to reveal the previous version.

Secure S3 Hosting

Recommended CloudFront Architecture

Browser

Connects to CloudFront over HTTPS, not directly to a public S3 bucket.

CloudFront + OAC

Verifies requests, caches static content, and obtains permission to access S3.

Private S3 Bucket

Keeps Block Public Access ON and allows only the CloudFront distribution to read objects.

Hands-on

Classroom Validation and Cleanup

Verify

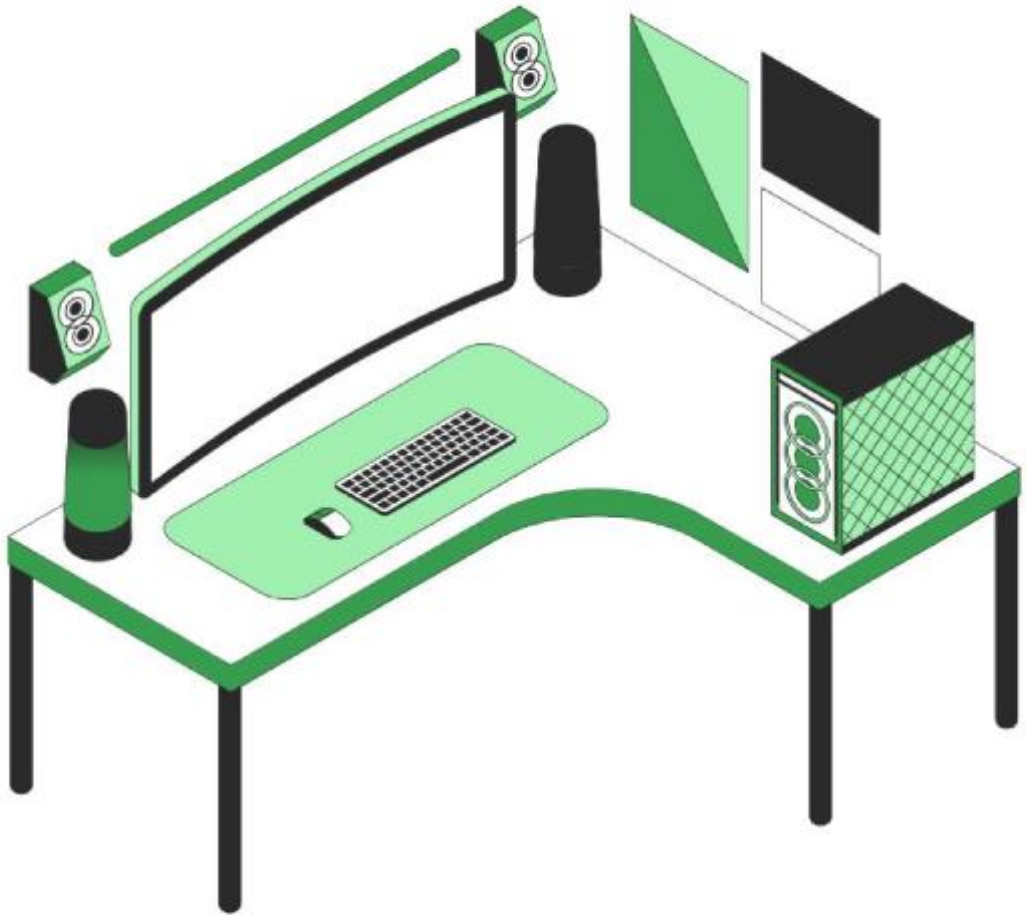
Check bucket permissions, website files, version IDs, and a temporary presigned URL.

Troubleshoot

403 Access Denied: investigate policies, OAC, object key and Block Public Access.

Cleanup

Empty versioned buckets including versions and markers; delete distributions and buckets; review billing.



Do you have any questions?

Send it to us! We hope you learned something new.

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